

1. The Binomial and Poisson distributions

What students need to learn:

The Binomial and Poisson distributions.

Students will be expected to use these distributions to model a real-world situation and to comment critically on their appropriateness. Cumulative probabilities by calculation or by reference to tables.

Students will be expected to use the additive property of the Poisson distribution – e.g. if the number of events per minute $\sim \text{Po}(\lambda)$ then the number of events per 5 minutes $\sim \text{Po}(5\lambda)$.

The mean and variance of the Binomial and Poisson distributions.

No derivations will be required.

The use of the Poisson distribution as an approximation to the Binomial distribution.

Questions may be set in the context of a hypothesis test.
